

UL' YA, Russian Federation

WMO: 310870

Lat: 58.833N

Lon: 141.850E

Elev: 5

StdP: 101.26

Time Zone: 10.00 (E10)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.2	(c) -27.7	(d) -37.4	(e) 0.1	(f) -28.8	(g) -35.3	(h) 0.1	(i) -26.9	(j) 18.4	(k) -21.9	(l) 17.8	(m) -19.8	(n) 9.0	(o) 320	(p) 0.852

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.8	(c) 20.4	(d) 13.8	(e) 18.2	(f) 13.5	(g) 16.6	(h) 13.2	(i) 15.5	(j) 18.1	(k) 14.6	(l) 16.7	(m) 13.9	(n) 15.8	(o) 4.5	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	10.3	16.2	13.7	9.8	15.3	13.1	9.4	14.7	43.6	18.3	41.1	16.8	39.2	15.8	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
16.4	14.3	12.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-31.2	26.2	2.7	4.3	-33.2	29.2	-34.7	31.7	-36.2	34.1	-38.2	37.2
			-31.4	18.0	2.7	2.3	-33.3	19.6	-34.9	21.0	-36.4	22.3	-38.3	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.4	-20.1	-17.3	-11.4	-4.2	2.0	7.8	12.1	13.0	8.2	-0.9	-12.7	-17.9		
	DBStd	12.44	5.10	4.95	4.83	3.99	3.07	2.97	2.49	2.57	3.34	4.68	5.85	5.61		
	HDD10.0	5084	933	766	663	425	248	81	8	5	72	339	681	864		
	HDD18.3	7925	1192	999	922	675	506	317	195	167	303	597	931	1122		
	CDD10.0	203	0	0	0	0	1	14	72	98	19	0	0	0		
	CDD18.3	2	0	0	0	0	0	0	1	2	0	0	0	0		
	CDH23.3	25	0	0	0	0	0	2	13	9	0	0	0	0		
	CDH26.7	7	0	0	0	0	0	0	6	1	0	0	0	0		
(14) Wind	WSAvg	4.8	7.7	6.2	4.0	3.4	3.3	3.0	2.9	3.0	3.8	5.1	7.6	8.1		
(15) Precipitation	PrecAvg	649	24	12	23	37	49	62	88	88	95	84	58	29		
	PrecMax	867	65	37	95	83	142	109	162	221	194	174	212	125		
	PrecMin	378	2	2	2	3	8	15	29	32	30	18	11	2		
	PrecStd	119	18	8	20	21	26	24	36	44	50	38	46	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.0	-2.9	0.8	7.0	15.8	21.5	23.2	24.5	19.1	11.1	3.1	-2.1		
		MCWB	-3.7	-3.3	-2.6	1.7	7.9	12.5	15.7	15.2	12.3	5.2	2.6	-2.8		
	2%	DB	-6.9	-5.2	-1.0	4.2	11.9	16.8	19.5	21.1	16.2	8.5	0.1	-3.6		
		MCWB	-8.1	-6.1	-3.8	0.1	6.5	11.8	14.3	14.0	11.9	4.6	-0.5	-3.9		
	5%	DB	-10.1	-7.9	-2.2	2.5	9.3	14.4	17.3	18.8	14.7	6.8	-1.3	-6.3		
		MCWB	-11.1	-8.9	-4.3	-0.4	5.0	10.6	13.6	13.7	11.7	3.9	-1.8	-7.0		
	10%	DB	-12.8	-10.4	-3.8	1.0	6.8	12.4	15.8	17.2	13.6	5.3	-3.6	-10.2		
		MCWB	-13.8	-11.8	-5.6	-1.3	3.9	9.9	13.4	13.5	11.2	3.0	-4.4	-11.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.6	-3.3	-1.7	2.2	8.8	14.0	17.0	14.7	7.3	2.7	-2.5			
		MCDB	-3.2	-2.9	-0.4	5.2	13.5	18.3	21.1	20.8	16.9	8.6	3.1	-2.3		
	2%	WB	-8.1	-5.9	-2.9	1.0	7.0	12.4	15.5	15.7	13.3	5.7	-0.2	-4.0		
		MCDB	-7.3	-5.4	-1.4	3.4	11.0	15.9	18.2	18.2	14.8	7.3	0.1	-3.6		
	5%	WB	-11.1	-8.9	-4.3	-0.1	5.5	11.2	14.4	15.0	12.6	4.5	-1.8	-7.0		
		MCDB	-10.1	-8.0	-2.6	1.5	8.5	13.6	16.3	17.1	13.9	6.2	-1.4	-6.5		
	10%	WB	-13.9	-11.8	-5.6	-0.8	4.3	10.2	13.7	14.3	11.7	3.4	-4.2	-11.4		
		MCDB	-13.0	-10.5	-3.9	0.5	6.3	11.8	15.2	16.2	13.1	5.0	-3.5	-10.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.5	8.4	7.3	5.5	5.8	5.6	6.8	7.5	5.9	4.6	4.1		
		MCDBR	6.2	6.5	8.9	8.5	11.1	10.4	9.4	10.5	8.9	7.9	4.6	5.5		
		MCWBR	6.1	6.1	7.1	6.1	6.8	6.3	6.0	6.4	6.3	6.0	4.8	5.7		
	5% WB	MCDBR	6.3	6.3	8.4	6.9	9.5	8.8	7.7	8.0	7.3	6.7	4.7	5.7		
		MCWBR	6.2	6.1	7.2	5.5	6.3	5.8	5.3	5.8	5.7	5.6	4.9	5.9		
(40) Clear-Sky Solar Irradiance	taub	0.232	0.268	0.326	0.431	0.460	0.435	0.488	0.428	0.350	0.322	0.264	0.221			
	taud	2.099	2.131	1.998	1.776	1.923	2.203	2.075	2.261	2.446	2.283	2.140	2.039			
	Ebn at Noon	594	751	791	750	765	801	740	765	786	676	532	462			
	Edh at Noon	48	78	122	183	170	131	146	113	80	69	46	35			
(44) All-Sky Solar Radiation	RadAvg	0.60	1.69	3.44	5.01	4.97	5.16	4.53	3.83	2.79	1.64	0.73	0.33			
	RadStd	0.04	0.08	0.17	0.38	0.58	0.50	0.73	0.48	0.36	0.19	0.09	0.03			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page